Problem Solving with Ratios

Exam code: 0607

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Ratio & Proportion

What type of ratio problems could I be asked to solve?

- **Simple ratio problems** are discussed in earlier revision notes, including
  - Writing ratios
  - The link between ratios and fractions
  - Equivalent ratios
  - Simplifying ratios
  - Sharing an amount in a given ratio
- **Further problems** involving ratio include
  - Ratios where you are given the **difference between** the **two parts**
    - E.g. Kerry is given $30 more than Kacey who is given $50
  - Ratios where **one quantity** is **given** and you have to **find** the **other quantity**
    - E.g. Kerry and Kacey are sharing money in the ratio 8 : 5, Kacey gets $50
  - Situations where you are given **two separate (two-part) ratios** but can combine them in to **one (three-part) ratio**
    - E.g. Kerry and Kacey are sharing money in the ratio 8 : 5 whilst Kacey is also sharing money with Kylie in the ratio 1 : 2

How do I solve a ratio problem when given the difference between two parts?

- **Find** the **difference in the number of parts** between the two quantities in the ratio
- **Compare** the difference in the **number of parts** with the difference between the **actual numbers**
- **Simplify** to find out the **value of one part**
- **Multiply** the **value of one part** by the **number of parts** for each quantity in the ratio
- **Multiply** the **value of one part** by the **total number of parts** to find the **total amount**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- **Compare** the **given quantity** with the relevant **number of parts** in the ratio
- **Simplify** to find the **value of one part**
- **Multiply** the **value of one part** by the **number of parts** in the remaining quantity in the ratio
- **Multiply** the **value of one part** by the **total number of parts** to find the **total amount**

How do I combine two ratios to make a three-part ratio?

- **Identify** the **link** between the two different ratios
- **Find equivalent ratios** for both original ratios, where the **value** of the **link** is the **same**
- **Join** the two, two-part ratios into a **three-part ratio**

Worked Example

(a) The ratio of cabbage leaves eaten by two rabbits, Alfred and Bob, is 7 : 3 respectively. It is known that Alfred eats 12 more cabbage leaves than Bob for a particular period of time.

Find the total number of cabbage leaves eaten by the rabbits and the number that each rabbit eats individually.

Answer:

The difference in the number of parts is

$$7 - 3 = 4 \text{ parts}$$

This means that

$$4 \text{ parts} = 12 \text{ cabbage leaves}$$

Dividing both by 4

$$1 \text{ part} = 3 \text{ cabbage leaves}$$

Find the total number of parts

$$7 + 3 = 10 \text{ parts}$$

Find the total number of cabbage leaves

$$10 \times 3 = 30$$

30 cabbage leaves in total

Find the number eaten by Alfred

$$7 \times 3 = 21$$

21 cabbage leaves

Find the number eaten by Bob

$$3 \times 3 = 9$$

9 cabbage leaves

Mark already has 36 litres of red paint, but no white paint. Calculate the volume of white paint that Mark needs to purchase in order to use all of his red paint, and calculate the total amount of pink paint this will produce.

Answer:

The ratio of red to white is

$$3 : 2$$

Mark already has 36 litres of red, so

$$36 \text{ litres} = 3 \text{ parts}$$

Divide both sides by 3

$$12 \text{ litres} = 1 \text{ part}$$

The ratio was 3 : 2
Find the volume of white paint, 2 parts

$$2 \times 12 = 24$$

24 litres of white paint

In total there are 5 parts, so the total volume of paint will be

$$5 \times 12 = 60$$

60 litres in total

(c) In Jamie's sock drawer the ratio of black socks to striped socks is 5 : 2 respectively. The ratio of striped socks to white socks in the drawer is 6 : 7 respectively.

Calculate the percentage of socks in the drawer that are black.

Answer:

Write down the ratios

$$B : S = 5 : 2$$
$$S : W = 6 : 7$$

S features in both ratios, so we can use it as a link
Multiply the B : S ratio by 3 to find an equivalent ratio
Both ratios are now comparing to 6 striped socks

$$B : S = 15 : 6$$
$$S : W = 6 : 7$$

Link them together

$$B : S : W = 15 : 6 : 7$$

Find the total number of parts

$$15 + 6 + 7 = 28$$

This means 15 out of 28 socks are black
Find 15 out of 28 as a decimal by completing the division

$$\frac{15}{28} = 0.535\ 714\ 285 \ldots$$

53.6 % of the socks are black

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